Winter Coordination Arlanda

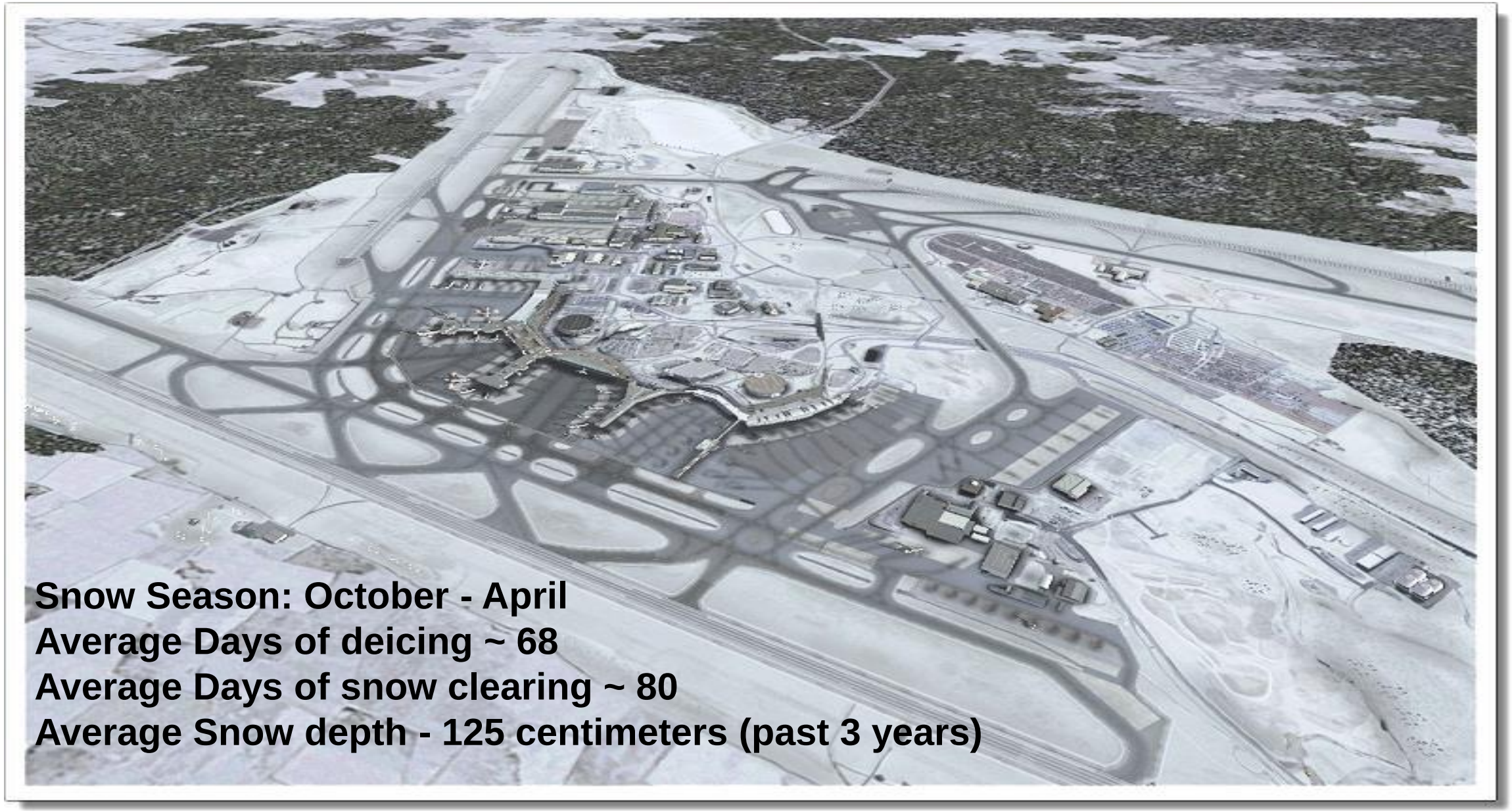
Daniel Skoglund
APOC Performance Team



Swedavia Airports

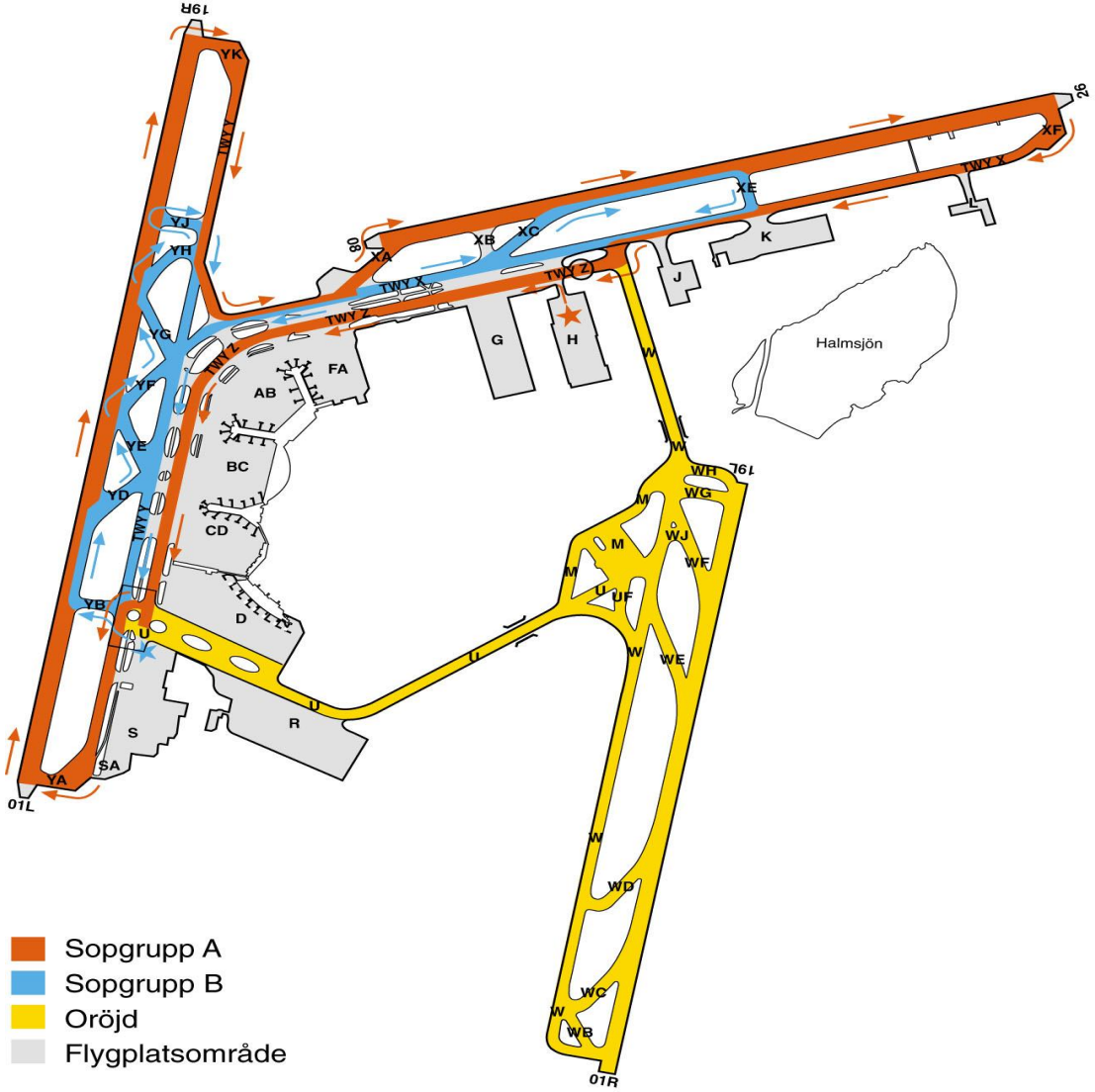


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Previous operating method

- Pre-defined cycles for snow removal of runways and taxiways is defined in a “Snowplan”.
- ”First come; first served” principle for snow removal of stand areas
- Standards/Goals
  - Runway system, serviceable condition H24
  - Friction value never less then 0.25
  - Time RWY 01L/19R = 8 minutes
  - Time RWY 08/26 = 6 minutes



Previous challenges

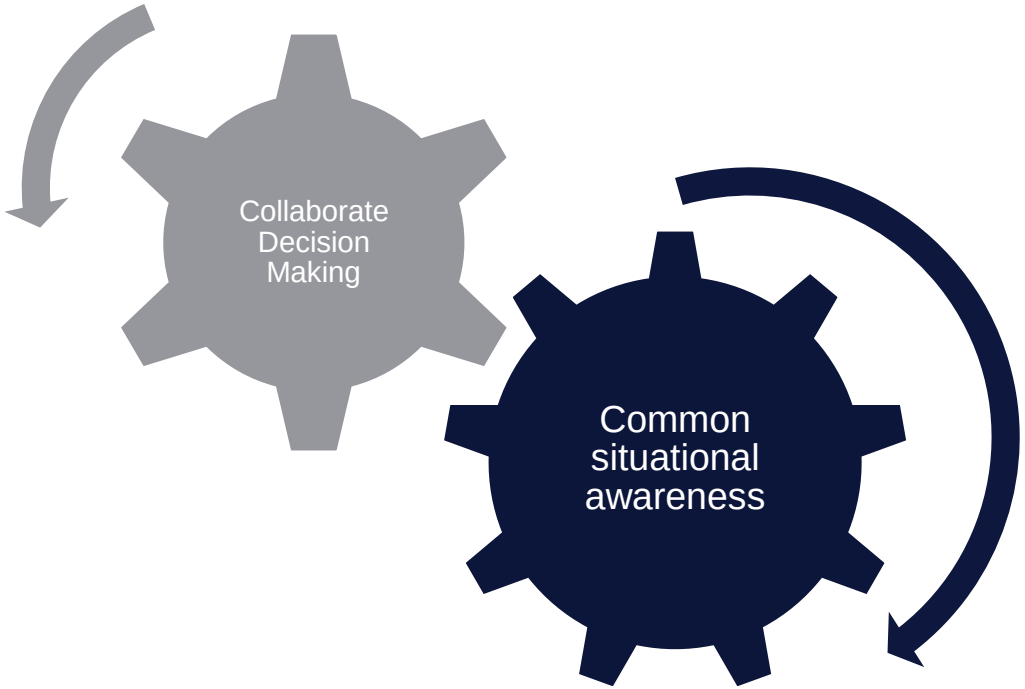
- Who determines the capacity (rate)?
  - ATC, Snow Removal, APOC?
- Coordination of snow removal at ramp areas
  - Since this was performed based on request it caused delays
- The snow removal method was not that flexible
  - Ad-hoc changes was difficult to solve since they followed a standardized method

New Operating Method



APOC mission

APOC enables common situational awareness which is our foundation for collaborate decision making relevant to optimize ARN’s capacity



Swedavia Airports

Collaboration by using APOC is key

Common Situation Awareness

- One common plan for the whole winter (Snowplan)
- One common weather platform provide by our MET provider
- One dedicated Winter Coordinator who liaisons with
  - APOC
  - ATC (Local TWR)
  - Snow Removal team
  - Ground Handlers
- Dual weekly weather coordination meeting
  - APOC
  - ATC (Local TWR and Approach control)
  - Met Provider



Collaborative Decision Making

- In case a negative weather situation might occur (Heavy snowfall), APOC Supervisor calls for a Tactical Traffic Forum
  - APOC
  - ATC (Local TWR and Approach control)
  - Met Provider
  - Ground Handlers
  - Airlines
- At this meeting all actors shares their demand and capacity, based on the balance/imbalance the forum creates a common plan to handle the expected capacity drop
- The plan is shared to all actors through the AOP

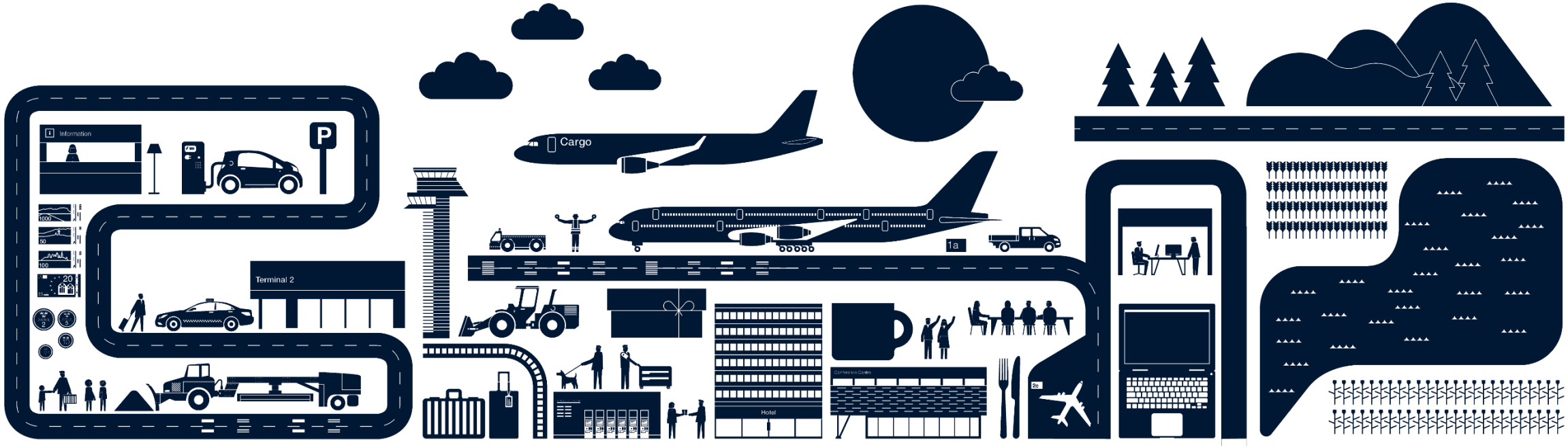
Joint table to calculate the rate

Number are not updated

RWY	Cap	B/A 1x M-G 2x P-M	Snow 60min x1, 45min x2, 30min x3	Apron 1 x per apron	Gate 1 x per terminal	De-ice 1 x per handler	TWY 1 x per clousre	Tech 1 x per system	Total
26/19R	61	-5 -10	-10 -15 -20	-6	-6	-6	-6	-6	
26/01L	50	-5 -10	-8 -12 -16	-5	-5	-5	-5	-5	
01L/08	68	-5 -10	-11 -17 -22	-7	-7	-7	-7	-7	
19R/08	72	-5 -10	-12 -18 -24	-7	-7	-7	-7	-7	
Single-RWY	40	-5 -10	-7 -11 -15	-4	-4	-4	-4	-4	
01L/01R 19R/19L	84	-5 -10	-14 -21 -28	-8	-8	-8	-8	-8	

Benefits from the new operating method

- Integrates all actors in the decision making process
  - Makes all the information available to all actors (Common situational awareness)
  - “If you have the information, you have to take responsibility”
- Publish clearly defined standards and priorities (Snowplan)
- Demand is driving the capacity


Information
P
Terminal 2
Cargo
1a
Hotel